

WILP, BITS Pilani

Session 13: Time Series Forecasting Methods

Course: Introduction to Statistical Methods (ISM) | Instructor: Dr. YVK Ravi Kumar

1 Moving Averages (Simple & Weighted)

Smooth out short-term fluctuations by averaging nearby historical observations.

Moving averages assume that future observations will resemble recent historical levels. Simple Moving Averages (SMA) weigh all k recent periods equally, whereas Weighted Moving Averages (WMA) assign larger weights to more recent observations.

Everyday Picture & Intuition

A simple moving average is like taking the average temperature over the last 3 days to guess tomorrow's weather. A weighted moving average gives yesterday 50% weight, two days ago 30%, and three days ago 20%.

1.1 Mathematical Formulas

$$\text{SMA}_t = \frac{y_{t-1} + y_{t-2} + \cdots + y_{t-k}}{k} \quad (1)$$

$$\text{WMA}_t = \frac{w_1 y_{t-1} + w_2 y_{t-2} + \cdots + w_k y_{t-k}}{\sum w_i} \quad (2)$$

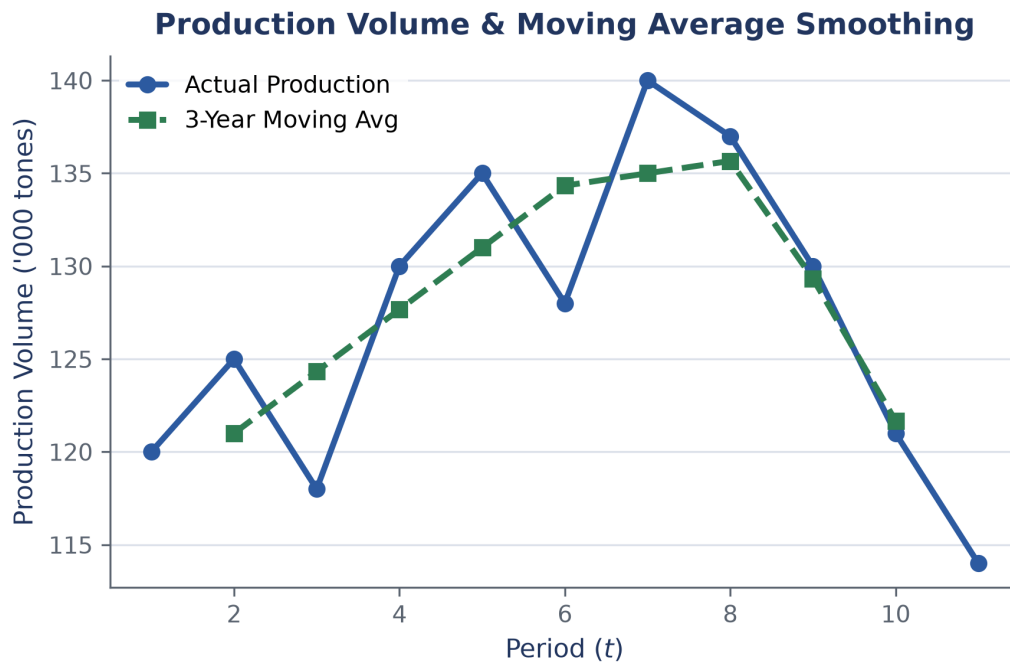


Figure 1: Comparison of raw production data against a 3-period moving average curve.

Fully Worked Example

Demand observations for recent months: Month 49 = 130, Month 50 = 128, Month 51 = 137.

Step 1: Compute 3-month Simple Moving Average forecast for Month 52:

$$F_{52} = \frac{137 + 128 + 130}{3} = \frac{395}{3} = 131.67 \quad (3)$$

Step 2: Compute Weighted Moving Average with weights $w = [3, 2, 1]$ (highest weight on most recent period):

$$F_{52} = \frac{3(137) + 2(128) + 1(130)}{3 + 2 + 1} = \frac{411 + 256 + 130}{6} = \frac{797}{6} = 132.83 \quad (4)$$

2 Simple Exponential Smoothing

Assign exponentially decaying weights to past historical observations.

Simple Exponential Smoothing (SES) updates forecasts using a smoothing constant $\alpha \in [0, 1]$:

$$F_{t+1} = \alpha y_t + (1 - \alpha)F_t = F_t + \alpha(y_t - F_t) \quad (5)$$

Everyday Picture & Intuition

Exponential smoothing is like updating your guess for commute time. Your new guess equals your old guess plus a fraction α of your error from today's commute.

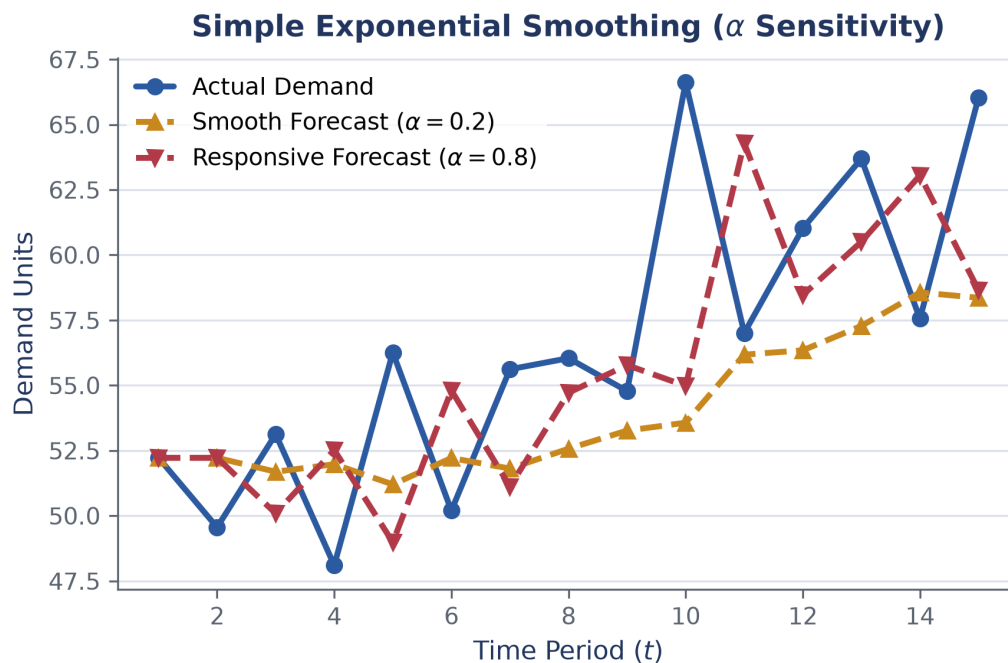


Figure 2: Sensitivity of exponential smoothing forecasts under different smoothing constants $\alpha = 0.2$ vs $\alpha = 0.8$.

Fully Worked Example

Suppose forecast for the first week of March was $F_1 = 500$ units, but actual demand was $y_1 = 450$ units. Let $\alpha = 0.3$.

Step 1: Calculate forecast for week 2:

$$F_2 = 0.3(450) + (1 - 0.3)(500) = 135 + 350 = 485 \quad (6)$$

Step 2: Suppose actual demand in week 2 is $y_2 = 505$ units. Calculate forecast for week 3:

$$F_3 = 0.3(505) + 0.7(485) = 151.5 + 339.5 = 491.0 \quad (7)$$

3 Holt's Double Exponential Smoothing

Track both baseline level and linear trend simultaneously in trending series.

When data exhibits a persistent linear trend, simple exponential smoothing lags behind. Holt's Double Exponential Smoothing introduces a second smoothing equation to update the trend component T_t alongside level L_t .

Everyday Picture & Intuition

If sales grow by 10 units every month, simple smoothing will always underestimate next month's sales. Holt's method tracks both the current height and the speed of growth.

3.1 Holt's Equations

$$L_t = \alpha y_t + (1 - \alpha)(L_{t-1} + T_{t-1}) \quad (8)$$

$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \quad (9)$$

$$\hat{y}_{t+h} = L_t + hT_t \quad (10)$$

Watch Out! Pitfalls

Simple moving averages and single exponential smoothing are only suitable for stationary series. For trending data, use Holt's method or ARIMA.

Key Takeaways

- Simple Moving Averages average recent k values equally; WMAs emphasize recent data.
- Exponential Smoothing updates forecasts by adding a fraction α of past forecast error.
- Holt's Double Exponential Smoothing incorporates a trend factor to eliminate lag on trending data.

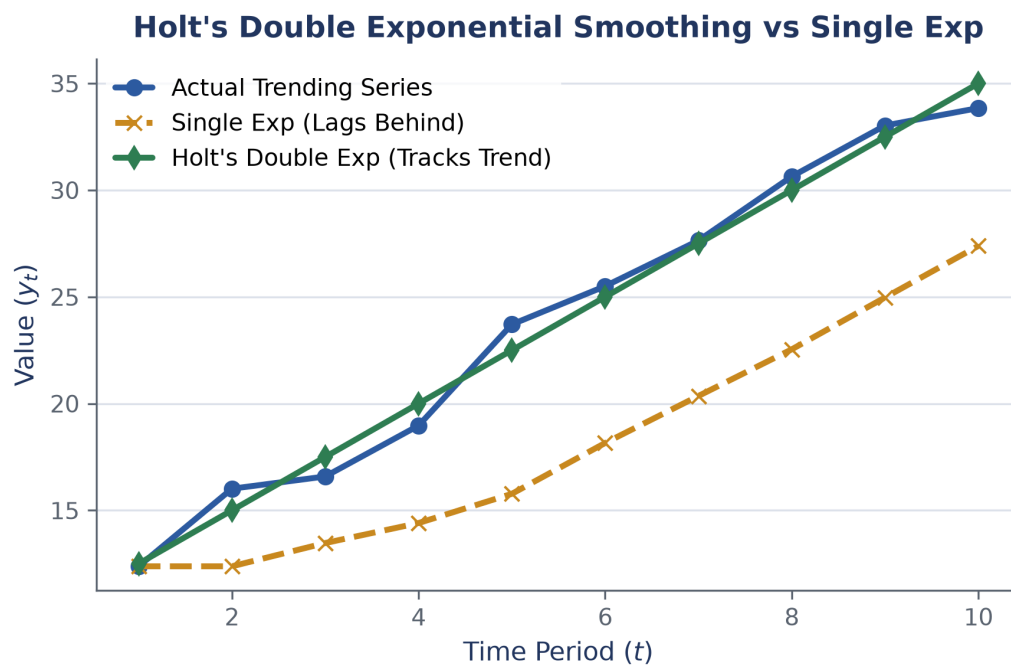


Figure 3: Holt's double exponential smoothing tracking trending time series without lagging.